A Pointwise Digit Certificate and a Cyclic-Modulus Counterexample

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Abstract

An explicit whole-block rotation is a square in every local field associated with the full cyclic modulus and the decimal base, but is not an integer square. The same rotation is locally a cube at all these primes without being an integer cube. This disproves the sufficiency of these congruence tests at every precision. Real-interval tests with the complete roots of actual decimal suffixes exclude the candidate: one added middle digit eliminates four square completions, and another eliminates four cube completions. Kashihara's conjecture remains open.

Write $W_j = 1\|\cdots\|j$, $D(j) = \mathrm{digits}(W_j)$, and $W_0 = D(0) = 0$. Moving the first k whole integer blocks to the end gives

$$R_{m,k} = 10^{D(k)}W_m - (10^{D(m)} - 1)W_k, \qquad 0 \le k < m. \tag{1}$$

The previous bounds $O(X)$ for square pairs and $O(X^2/\log X)$ for perfect-power pairs permit isolated exceptions. No counting bound is used below. All integer positions in (1), including the middle blocks, retain their prescribed values.

1 A complete cyclic modulus does not suffice

Proposition 1.1. *The actual rotation*

$$R = R_{16,1} = 23456789101112131415161, \qquad \Delta = 10^{23} - 1,$$

is a square in $\mathbb{Q}_p$ for every prime $p \mid 10\Delta$, but is not an integer square. Consequently it is a square modulo $10^a\Delta^b$ for every pair of nonnegative integers a, b. Thus even the complete cyclic modulus together with arbitrarily precise decimal congruences cannot reject every individual rotation.

Proof. The word is $2\|3\|\cdots\|16\|1$, with 23 digits, and (1) reads $R = 10W_{16} - \Delta$. Direct integer arithmetic gives

$$r = 24515341518503348803133, \qquad r^2 \equiv R \pmod{\Delta}, \qquad \gcd(2r, \Delta) = 1.$$

For every prime divisor p of Δ, this is a nonsingular root modulo p, which lifts to all p-adic precisions by Hensel's lemma. No factorization of Δ is needed. Also $R \equiv 1 \pmod 8$ and $R \equiv 1 \pmod 5$, giving square roots in $\mathbb{Q}_2$ and $\mathbb{Q}_5$. These are integral local roots. The Chinese remainder theorem proves the assertion for $10^a\Delta^b$. Section 3 proves that R is not an integer square using an actual middle digit. □

This is a counterexample to the sufficiency of these local conditions, not to Kashihara's conjecture. In particular, retaining all middle digits in a congruence is different from forcing a root of the required real size. No claim that every pointwise method fails follows.

2 Exact compatibility with the real interval

Let a positive integer R have E digits. Prescribe its first h digits P and its last q digits $U = R \bmod 10^q$, with $q \ge 3$. The digits of U are taken from the actual word, and may extend past the original moved prefix into its middle blocks. Put

$$A = \left\lceil \sqrt{P10^{E-h}} \right\rceil, \qquad B = \left\lfloor \sqrt{(P+1)10^{E-h} - 1} \right\rfloor.$$

Assume $v_2(U), v_5(U) < q$. An odd valuation excludes squares immediately. Otherwise write $v_2(U) = 2a$, $v_5(U) = 2b$, and

$$z = 2^a5^b, \qquad H = 10^q/z.$$

Solving the two unit congruences at 2 and 5 and applying CRT gives a complete set $\mathcal{A} \subset [0, H)$ of at most eight root classes:

$$y^2 \equiv U \pmod{10^q} \iff y \equiv r \pmod{H} \text{ for some } r \in \mathcal{A}. \tag{2}$$

Indeed, division by p^{2a} or p^{2b} gives unit square roots, at most four at 2 and two at 5. Multiplying back gives root moduli 2^{q-a} and 5^{q-b}, whose product is H.

Lemma 2.1 (Pointwise interval test)**.** *A square with the prescribed length, prefix and suffix exists if and only if, for some* $r \in \mathcal{A}$*,*

$$\max\left(0, \left\lceil \frac{A-r}{H} \right\rceil\right) \;\le\; \left\lfloor \frac{B-r}{H} \right\rfloor.$$

If $B - A < H$*, at most eight possible positive roots occur. Failure of all these inequalities excludes the particular rotation supplying the digits.*

Proof. The prefix and length are equivalent to $A \le y \le B$; the suffix is equivalent to (2). Write $y = r + jH$, with $j \ge 0$, and solve the two interval inequalities for j. An interval shorter than H contains at most one integer from each root class. □

The test is exact for the selected windows. If it leaves roots, the additional actual middle digits must be checked; their values cannot be filled freely. Increasing q preserves every genuine square candidate. At $q = E$, the test is an exact square decision, since $y^2 < 10^E$ and $y^2 \equiv R \pmod{10^E}$ force equality. This termination observation does not prove uniform exclusion for the infinite rotation family.

3 One middle digit excludes the isolated candidate

For Proposition 1.1, the first digit is 2, so

$$[A, B] = [141421356238,\ 173205080756].$$

At precision $q = 12$, the actual suffix is

$$U = 112131415161 = 1\|12\|13\|14\|15\|16\|1.$$

Its first displayed 1 is the final digit of the middle block 11; it is not part of the original moved prefix $W_1 = 1$. Here $H = 10^{12}$. The complete eight roots modulo H are

55164164619	104480570869	395519429131	444835835381
555164164619	604480570869	895519429131	944835835381.

Each entry squares to U modulo 10^{12}. They are distinct, and the unit-root bound of eight proves completeness. None belongs to $[A, B]$; also $B < H$, so no positive translate of a listed root can belong to the interval. Lemma 2.1 proves that R is not a square.

The extra middle digit is substantive. The weaker 11-digit suffix 12131415161 and the same first digit and total length admit the following four squares:

Root	Square	Twelfth digit from the right
144835835381	20977419210512131415161	5
145519429131	21175904254612131415161	6
154480570869	23864246776012131415161	0
155164164619	24075917981912131415161	9

All four have 23 digits, start with 2, and share the actual last 11 digits. The next actual digit is 1, whereas these completions have $5, 6, 0, 9$. The complete 12-digit calculation above excludes every completion, not merely these four examples.

4 The same obstruction and digit mechanism for cubes

Proposition 4.1. *The rotation in Proposition 1.1 is a cube in* $\mathbb{Q}_p$ *for every* $p \mid 10\Delta$*, and hence modulo* $10^a\Delta^b$ *for all* $a, b \ge 0$*, but it is not an integer cube.*

Proof. Write $\Delta = 9N$, where $N = 11111111111111111111$ is prime. For a self-contained primality certificate,

$$N - 1 = 2 \cdot 5 \cdot 11^2 \cdot 23 \cdot 4093 \cdot 8779 \cdot 21649 \cdot 513239.$$

Every displayed factor is prime by trial division. Direct modular exponentiation gives $11^{N-1} \equiv 1 \pmod{N}$ and

$$\gcd\left(11^{(N-1)/p} - 1, N\right) = 1 \quad (p = 2, 5, 11, 23, 4093, 8779, 21649, 513239).$$

Thus for every prime divisor r of N, the order of 11 modulo r is $N - 1$, forcing $r \ge N$ and proving primality. Since $N \equiv 2 \pmod 3$, cubing is bijective on $\mathbb{F}_N^\times$ and its roots lift by Hensel. Cubing is also bijective on $\mathbb{Z}_2^\times$ and $\mathbb{Z}_5^\times$. Finally $R \equiv 1 \pmod 9$, so R is a cube in $\mathbb{Z}_3^\times$: the 3-adic logarithm identifies cubing on $1 + 3\mathbb{Z}_3$ with multiplication by 3, whose image is $9\mathbb{Z}_3$. CRT proves the congruence assertion.

The first digit and length force a positive cube root into

$$[A_3, B_3] = [27144177,\ 31072325].$$

The actual seven-digit suffix is $1415161 = 14\|15\|16\|1$. Cubing is a bijection on the units modulo 10^7, and the suffix has the unique root 2393321. The consecutive representatives $22393321 < A_3$ and $32393321 > B_3$ exclude a cube root in this interval. At six digits there are exactly four roots in the same interval,

$$27393321, \quad 28393321, \quad 29393321, \quad 30393321.$$

Their cubes have seventh digits from the right $6, 9, 2, 5$, respectively. The actual seventh digit, the first digit of the middle block 14, is 1, and eliminates all four completions. □

More generally, let $e > 1$ be coprime to 10 and suppose $v_p(U) < q$ for $p = 2, 5$. Unless both valuations are divisible by e, no eth power is possible. Otherwise put $v_2(U) = ea$ and $v_5(U) = eb$. Powering by e is bijective in each of the two unit groups; inversion of e modulo the corresponding Euler totient finds its unique root. After rescaling and CRT the complete solution is one class modulo

$$H_e = 2^{q-(e-1)a}5^{q-(e-1)b}.$$

Lemma 2.1 applies with the real eth-root interval in place of $[A, B]$. This gives an exact pointwise test for every odd prime exponent except 5, without claiming that its interval is always empty.

For $e = 5$, the same valuation reduction applies. Fifth powering is bijective on the 2-adic unit group. A unit u modulo 5^n, $n \geq 2$, has fifth roots exactly when $u^4 \equiv 1 \pmod{25}$, and then has five roots; at $n = 1$ it has one root. This follows from the unit group $C_4 \times C_{5^{n-1}}$. To compute the five roots, start with the unique $r \in \{1, 2, 3, 4\}$ satisfying $r^5 \equiv u \pmod{25}$ and lift using the derivative $5r^4$; the five final roots differ by 5^{n-1}. Thus at most five CRT classes require the same interval test. The displayed rotation already fails modulo 25: $R \equiv 11$, whose fourth power is 16, not 1. The implementation therefore covers every prime exponent, subject to the stated suffix-valuation hypothesis.

5 Verification and the remaining problem

The independent Python verifier checks every displayed certificate exactly and compares the GMP/OpenMP implementation with the earlier Python square-root solver, independent decimal digit lifting for odd exponents, and small exhaustive intervals. It also checks the complete Lucas primality certificate used in Proposition 4.1. The window and Hensel/CRT tools are standard; no priority claim is made. No uniform theorem proves that the interval tests reject every rotation. Kashihara's conjecture remains open; the existing sparsity bounds are unchanged.